

Chapter 3

Measuring Business Income



Financial Accounting, 11e

Learning Objectives

- LO 1** Define net income and explain the assumptions underlying income measurement and their ethical application.
- LO 2** Define accrual accounting and explain how it is accomplished.
- LO 3** Identify the four situations that require adjusting entries and illustrate typical adjusting entries.
- LO 4** Prepare financial statements from an adjusted trial balance.
- LO 5** Explain and prepare closing entries.
- LO 6** Use accrual-based information to analyze cash flows.

Profitability Measurement Issues and Ethics

- Profit means different things to different people.
- Accountants prefer the term **net income**.
- Net Income
 - accumulated in the Retained Earnings account
 - reported on the income statement
 - used to assess a company's profitability
 - In its simplest form, net income is
 - **Net Income = Revenues – Expenses**
- **Net loss:** Expenses exceed revenues

Net Income

- **Revenues** are increases from selling goods, rendering services, or performing other business activities.
 - total of Accounts Receivable or Notes Receivable and the total cash received from customers
- **Expenses** are decreases in stockholders' equity resulting from:
 - The cost of selling goods or rendering services
 - The cost of activities necessary to carry on a business
 - salaries expense, rent expense, advertising expense, utilities expense, and depreciation
 - often called the *cost of doing business* or *expired costs*.
- Stockholders' investments increase stockholders' equity but are not revenues, and dividends decrease stockholders' equity but are not expenses.

Income Measurement Assumptions

- Justification for all the techniques of income measurement rests on the assumption of continuity.
 - **Continuity**: Measuring business income so that certain expense and revenue transactions are allocated over several accounting periods.
 - Example: Cost of certain assets held on balance sheet until a future accounting period, when it becomes an expense
 - **Going concern**: The assumption that the business will continue to operate indefinitely.

Periodicity

- **Periodicity**: Measuring business income such that expenses and revenues are assigned to a specific accounting period.
 - Example: A company purchases a building, estimates the number of years it will be used, then assigns a portion of the cost of the building to each period.
- Estimate the business's net income in accounting periods.
 - **Fiscal year**: 12-month accounting period, usually a calendar year, Jan 1 to Dec. 31
 - **Interim periods**: Accounting periods of less than a year

Matching Rule

(slide 1 of 2)

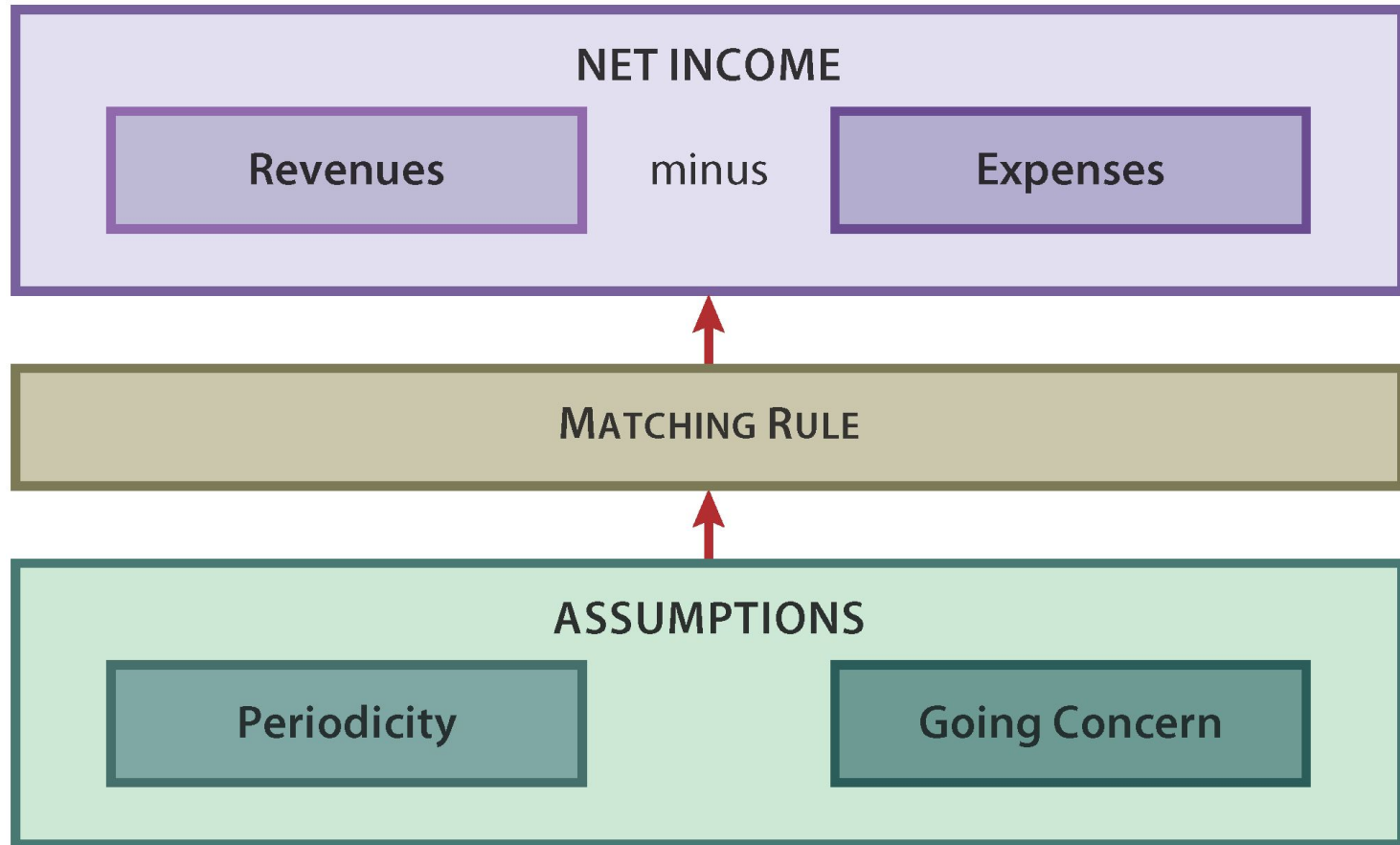
- **Matching rule:** Revenues and expenses assigned to the accounting period in which they occur.
 - Expenses assigned to the accounting period in which they are used to produce revenue.
 - Not always clear direct cause-and-effect relationship; costs allocated in a systematic way among the accounting periods
 - Example: A building's cost is expensed over the building's expected useful life.

Matching Rule

(slide 2 of 2)

- **Cash basis of accounting:** Accounting for revenues in the period in which cash is received and for expenses in the period in which cash is paid.
 - This method works well for some small businesses and many individuals but does not fit the needs of most businesses.

Assumptions and the Matching Rule



Ethics and the Matching Rule

- Matching rule involves making assumptions and exercising judgment.
 - Example: Estimating the useful life of a building
- **Earnings management**: Manipulation of revenues and expenses to achieve a specific outcome.
 - Meet a previously announced goal and thus meet the expectations of the market.
 - Keep the company's stock price from dropping.
 - Meet a goal that will enable it to earn bonuses.
 - Avoid embarrassment.

Earnings Management

- Outside a reasonable range, the financial statements become misleading.
- Preparation of intentionally misleading financial statements constitutes fraudulent financial reporting.
 - Example: **Dell Computer** had to restate four years of its financial results because senior executives improperly applied accrual accounting to give the impression that the company was meeting quarterly earnings targets.



Stop & Apply

Match the concepts on the right with the assumptions or actions on the left:

- | | |
|---|------------------------|
| 1. Increases in stockholders' equity resulting from selling goods, rendering services, or performing other business activities | a. Net income |
| 2. Manipulation of revenues and expenses to achieve a specific change in stockholders' equity | b. Revenues |
| 3. Increase in stockholders' equity that results from a company's operations | c. Expenses |
| 4. Decreases in stockholders' equity resulting from the cost of selling goods, rendering services, and performing other business activities | d. Earnings management |

SOLUTION 1. b; 2. d; 3. a; 4. c

Accrual Accounting

- **Accrual accounting:** The techniques accountants use to apply the matching rule.
- Accomplished in the following ways:
 - Recognizing revenues when they are earned.
 - Recognizing expenses when they are incurred.
 - Adjusting the accounts.

Recognizing Revenues

- **Revenue recognition:** The process of determining when revenue should be recorded.
- SEC requires that all the following conditions be met before revenue is recognized:
 - Persuasive evidence of an arrangement exists.
 - A product or service has been delivered.
 - The seller's price to the buyer is fixed or determinable.
 - Collectability is reasonably assured.

Recognizing Expenses

- Expenses are recorded when all of the following conditions have been met:
 - There is an agreement to purchase goods or services.
 - The goods have been delivered or the services rendered.
 - A price has been established or can be determined.
 - The goods or services have been used to produce revenue.

Adjusting the Accounts

- Adjustments are made because accounting periods end on a particular day.
 - Balance sheet must list all assets and liabilities as of the end of that day.
 - Income statement must contain all revenues and expenses applicable to the period ending on that day.

Trial Balance for Creative Designs, Inc.

Creative Designs, Inc.

Trial Balance

July 31, 2011

Cash	\$22,480	
Accounts Receivable	4,600	
Office Supplies	5,200	
Prepaid Rent	3,200	
Office Equipment	16,320	
Accounts Payable		\$ 6,280
Unearned Design Revenue		1,400
Common Stock		40,000
Dividends	2,800	
Design Revenue		12,400
Wages Expense	4,800	
Utilities Expense	680	
	<u>\$60,080</u>	<u>\$60,080</u>

Adjustments and Ethics

- SEC has identified accrual accounting and adjustments as an area with potential for abuse and misrepresentation.
- Adjustments affect performance measures of profitability and liquidity.
- Adjusting entries affect:
 - Net income on the income statement and profitability comparisons between accounting period
 - Assets and liabilities on the balance sheet and information about a company's future cash inflows and outflows.



Stop & Apply

Four conditions must be met before revenue should be recognized. Identify which of these conditions applies to the following actions of Hasting Corporation in reference to a client:

- a. Determines that the client has a good credit rating.
- b. Agrees to a price for services before it performs them.
- c. Performs services.
- d. Signs a contract to perform services.

SOLUTION

- a. Collectability is reasonably assured.
- b. The seller's price to the buyer is fixed or determinable.
- c. A product or service has been delivered.
- d. Persuasive evidence of an arrangement exists.

The Adjustment Process

- Transactions that span more than one accounting period require the use of **adjusting entries**.
 - **Deferral**: The postponement of the recognition of an expense already paid or of revenue received in advance.
 - **Accrual**: The recognition of a revenue or expense that has arisen but not been recorded during the accounting period.

The Four Types of Adjustments

		BALANCE SHEET	
		Asset	Liability
I N C O M E S T A T E M E N T	Expense	1. Allocating recorded costs between two or more accounting periods.	2. Recognizing unrecorded expenses.
	Revenue	4. Recognizing unrecorded, earned revenues.	3. Allocating recorded, unearned revenues between two or more accounting periods.

Type 1 Adjustment:

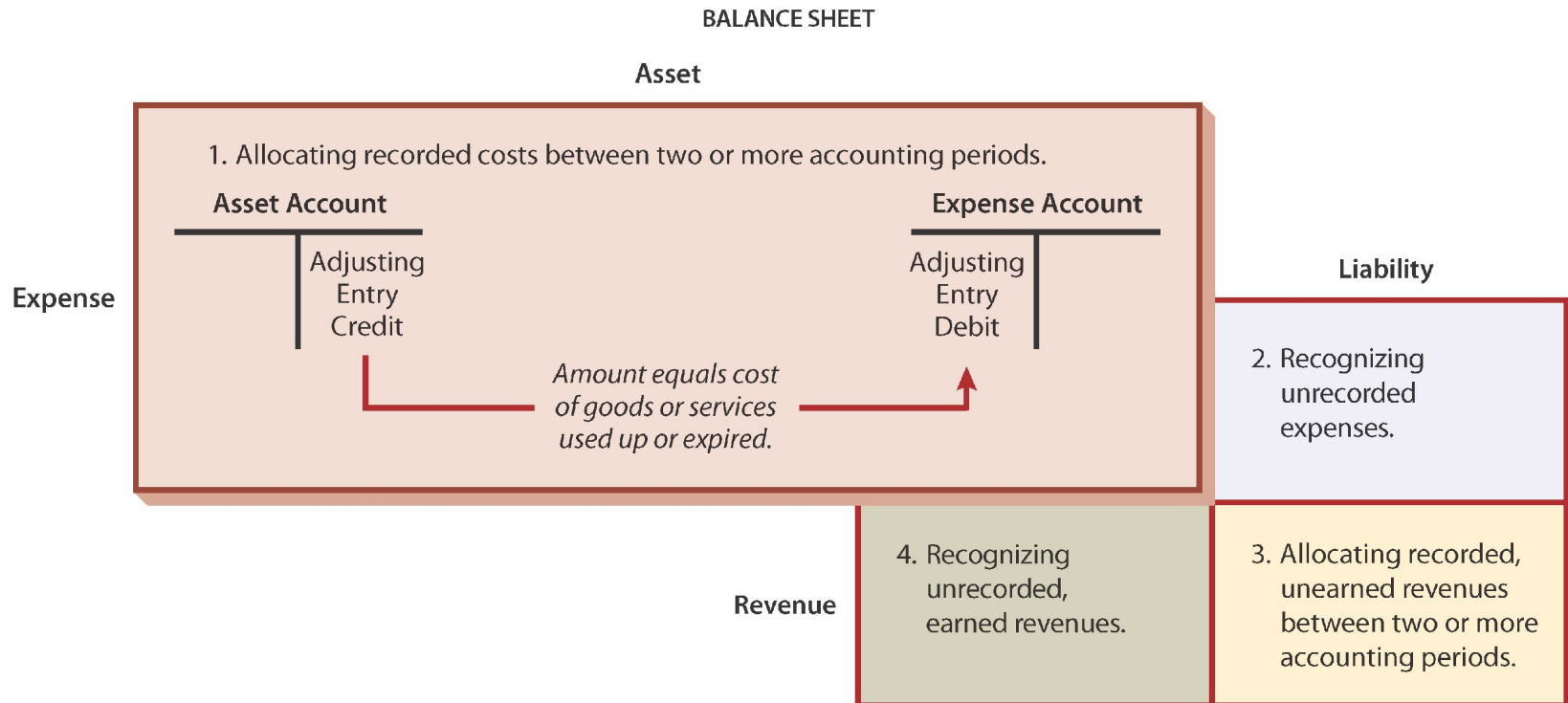
Allocating Recorded Costs (Deferred Expenses)

- **Prepaid expenses:** Expenses paid in advance.
 - Example: Rent, supplies, and insurance and depreciation of plant and equipment.
 - **Depreciation:** The amount of the cost of a long-term asset allocated to any one accounting period over its estimated useful life.
 - If adjusting entries for prepaid expenses are not made:
 - The company's assets will be overstated (stockholders' equity on the balance sheet will be overstated)
 - The company's expenses will be understated (net income on the income statement will be overstated)

Adjustment for Prepaid (Deferred) Expenses

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EXAMPLE:

Adjustment for Prepaid Rent (1 of 2)

- **July 31:** Expiration of one month's rent, \$1,600
- **Analysis:** Expiration of prepaid rent
 - ▼ decreases the asset account *Prepaid Rent* with a credit and
 - ▲ increases the expense account *Rent Expense* with a debit.

EXAMPLE:

Adjustment for Prepaid Rent (2 of 2)

Application of Double Entry:

Assets

PREPAID RENT

July 33,200

July 311,600

Bal.1,600

=

Liabilities

+

Stockholders' Equity

RENT EXPENSE

July 311,600

Journal Entry:

July 31

Rent Expense

Prepaid Rent

Dr.

Cr.

1,600

1,600

- **Comment:** The Prepaid Rent account now has a balance of \$1,600, which represents one month's rent that will be expensed during August. The logic in this analysis applies to all prepaid expenses.

EXAMPLE:

Adjustment for Supplies (1 of 2)

- **July 31:** Consumption of supplies, \$1,540
- **Analysis:** Consumption of office supplies
 - ▼ decreases the asset account *Office Supplies* with a credit and
 - ▲ increases the expense account *Office Supplies Expense* with a debit.

EXAMPLE:

Adjustment for Supplies (2 of 2)

Application of Double Entry:

Assets		=	Liabilities	+	Stockholders' Equity	
OFFICE SUPPLIES					OFFICE SUPPLIES EXPENSE	
July 5	5,200				July 31	1,540
Bal.	3,660					

Journal Entry:

		Dr.	Cr.
July 31	Office Supplies Expense	1,540	
	Office Supplies		1,540

- **Comment:** The asset account Office Supplies now reflects the correct balance of \$3,660 of supplies yet to be consumed. The logic in this example applies to all kinds of supplies.

Depreciation of Plant and Equipment

- To maintain historical cost in specific long-term asset accounts, separate accounts – **Accumulated Depreciation accounts** – are used to accumulate the depreciation on each long-term asset.
 - **Contra account:** A separate account paired with a related account.
 - Accumulated Depreciation accounts are contra accounts.
 - **Carrying value:** The amount in the asset account reduced by the related contra amount.
 - Over time, the accumulated depreciation grows, and the carrying value shown as an asset declines.

EXAMPLE: Adjustment for Plant and Equipment (1 of 3)

- *July 31:* Depreciation of office equipment, \$300
- *Analysis:* Depreciation
 - ▼ decreases the asset account *Office Equipment*
 - ▲ increases the contra account *Depreciation Expense—Office Equipment* with a credit,
 - ▲ and increases the expense account *Depreciation Expense—Office Equipment* with a debit.

EXAMPLE: Adjustment for Plant and Equipment (2 of 3)

Application of Double Entry:

Assets		=	Liabilities	+	Stockholders' Equity	
OFFICE EQUIPMENT					DEPRECIATION EXPENSE— OFFICE EQUIPMENT	
July 6	16,320				July 31	300
ACCUMULATED DEPRECIATION— OFFICE EQUIPMENT						
	July 31		300			

Journal Entry:

July 31	Depreciation Expense— Office Equipment	Dr. 300	Cr.
	Accumulated Depreciation— Office Equipment		300

EXAMPLE: Adjustment for Plant and Equipment (3 of 3)

- *Comment:* The carrying value of Office Equipment is \$16,020 (\$16,320 – \$300) and is presented on the balance sheet as follows:

PROPERTY, PLANT, AND EQUIPMENT

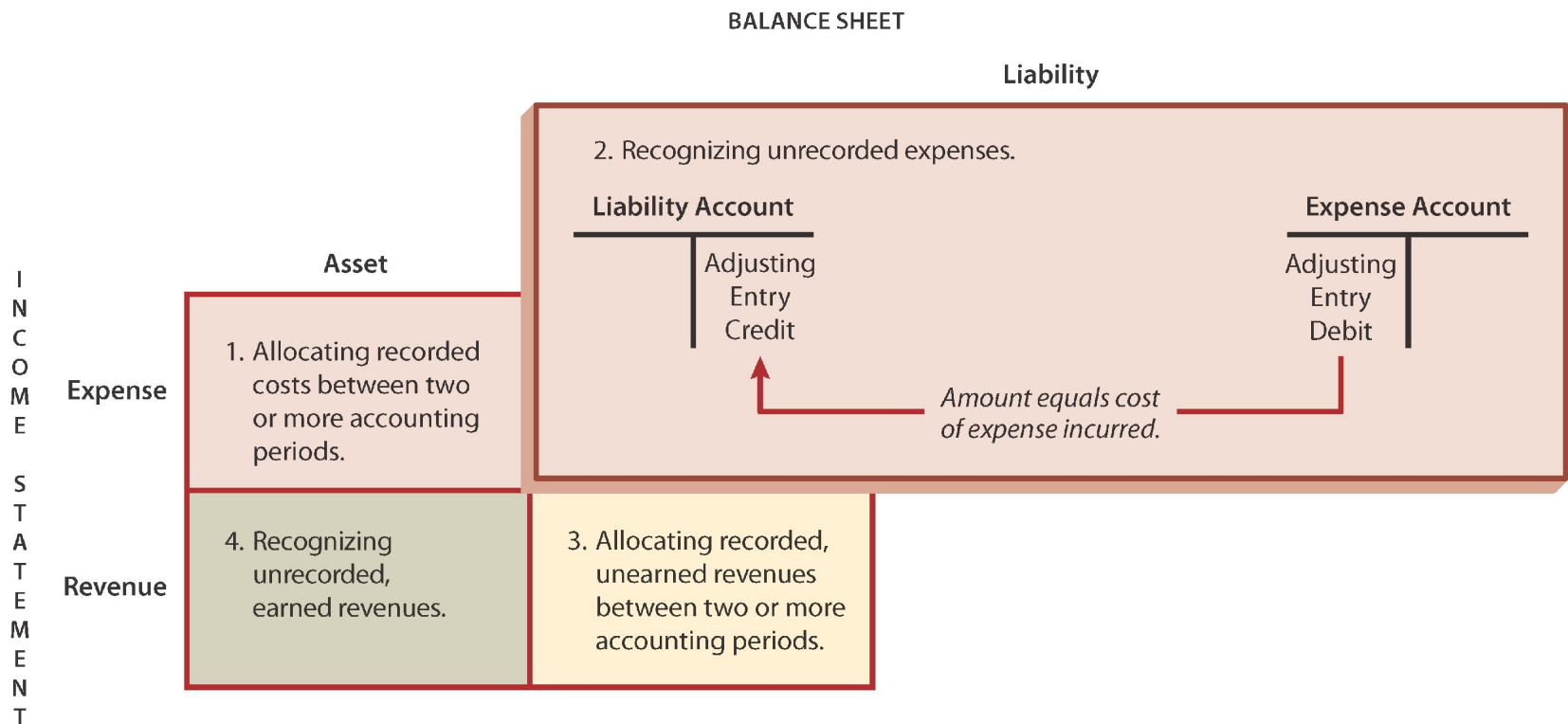
Office equipment \$16,320

Less accumulated depreciation 300 \$16,020

Type 2 Adjustment: Recognizing Unrecorded Expenses

- Expenses incurred during the period but not recorded in the accounts require adjusting entries.
 - Examples: interest on borrowed money, wages, taxes, and utilities
- As the expense and the corresponding liability accumulate, they are said to accrue—hence, the term **accrued expenses**.

Adjustment for Unrecorded (Accrued) Expenses



EXAMPLE: Adjustment for Unrecorded Wages (1 of 2)

- **July 31:** Accrual of unrecorded wages, \$720
- **Analysis:** Accrual of wages
 - ▲ *increases* the stockholders' equity account *Wages Expense* with a debit and
 - ▲ *increases* the liability account *Wages Payable* with a credit.

EXAMPLE: Adjustment for Unrecorded Wages (1 of 2)

Application of Double Entry:

Assets	=	Liabilities	+	Stockholders' Equity
		WAGES PAYABLE		WAGES EXPENSE
		July 31 720		July 26 4,800
				31 720
				Bal. 5,520

Journal Entry:

July 31 Wages Expense
Wages Payable

Dr. Cr.
720 720

- **Comment:** Note that the increase in Wages Expense will decrease stockholders' equity and that total wages for the month are \$5,520, of which \$720 will be paid next month.

EXAMPLE: Adjustment for Estimated Income Taxes (1 of 2)

- **July 31:** Accrual of estimated income taxes, \$800
- **Analysis:** Accrual of income taxes
 - ▲ *increases* the stockholders' equity account *Income Taxes Expense* with a debit and
 - ▲ *increases* the liability account *Income Taxes Payable* with a credit.

EXAMPLE: Adjustment for Estimated Income Taxes (1 of 2)

Application of Double Entry:

Assets	=	Liabilities	+	Stockholders' Equity
		INCOME TAXES PAYABLE		INCOME TAXES EXPENSE
		July 31 800		July 31 800

Journal Entry:

July 31 Income Taxes Expense
 Income Taxes Payable

Dr.	Cr.
800	
	800

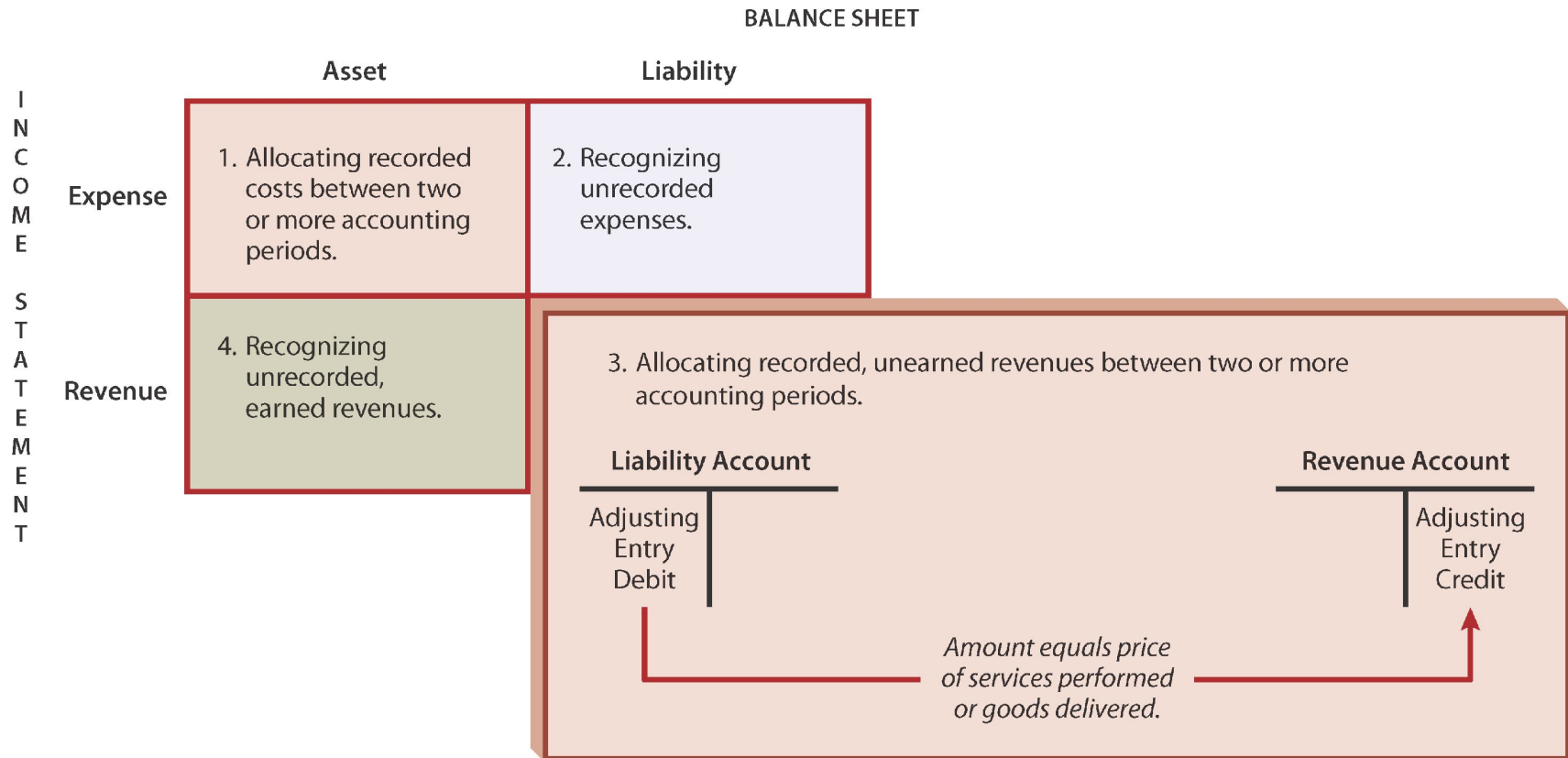
- **Comment:** Note that the increase in Income Taxes Expense will decrease stockholders' equity. There are many types of accrued expenses, and the adjustments made for all of them follow the same procedure as the one used for accrued wages and accrued income taxes.

Type 3 Adjustment:

Allocating Recorded, Unearned Revenues

- **Unearned revenues:** Revenues received before they are earned.
- When a company receives revenues in advance they are shown in a liability account.
 - Example: Payment in advance for magazine subscriptions

Adjustment for Unearned (Deferred) Revenues



EXAMPLE: Adjustment for Unearned Revenue (1 of 2)

- *July 31:* Performance of services paid for in advance, \$800
- *Analysis:* Performance of the services for which payment had been received in advance
 - ▲ *increases* the stockholders' equity account *Design Revenue* with a credit and
 - ▼ *decreases* the liability account *Unearned Design Revenue* with a debit.

EXAMPLE: Adjustment for Unearned Revenue (2 of 2)

Application of Double Entry:

Assets	=	Liabilities		+	Stockholders' Equity		
		UNEARNED DESIGN REVENUE			DESIGN REVENUE		
		July 31	800			July 10	2,800
						15	9,600
						31	800

Journal Entry:

	Dr.	Cr.
July 31 Unearned Design Revenue	800	
Design Revenue		800

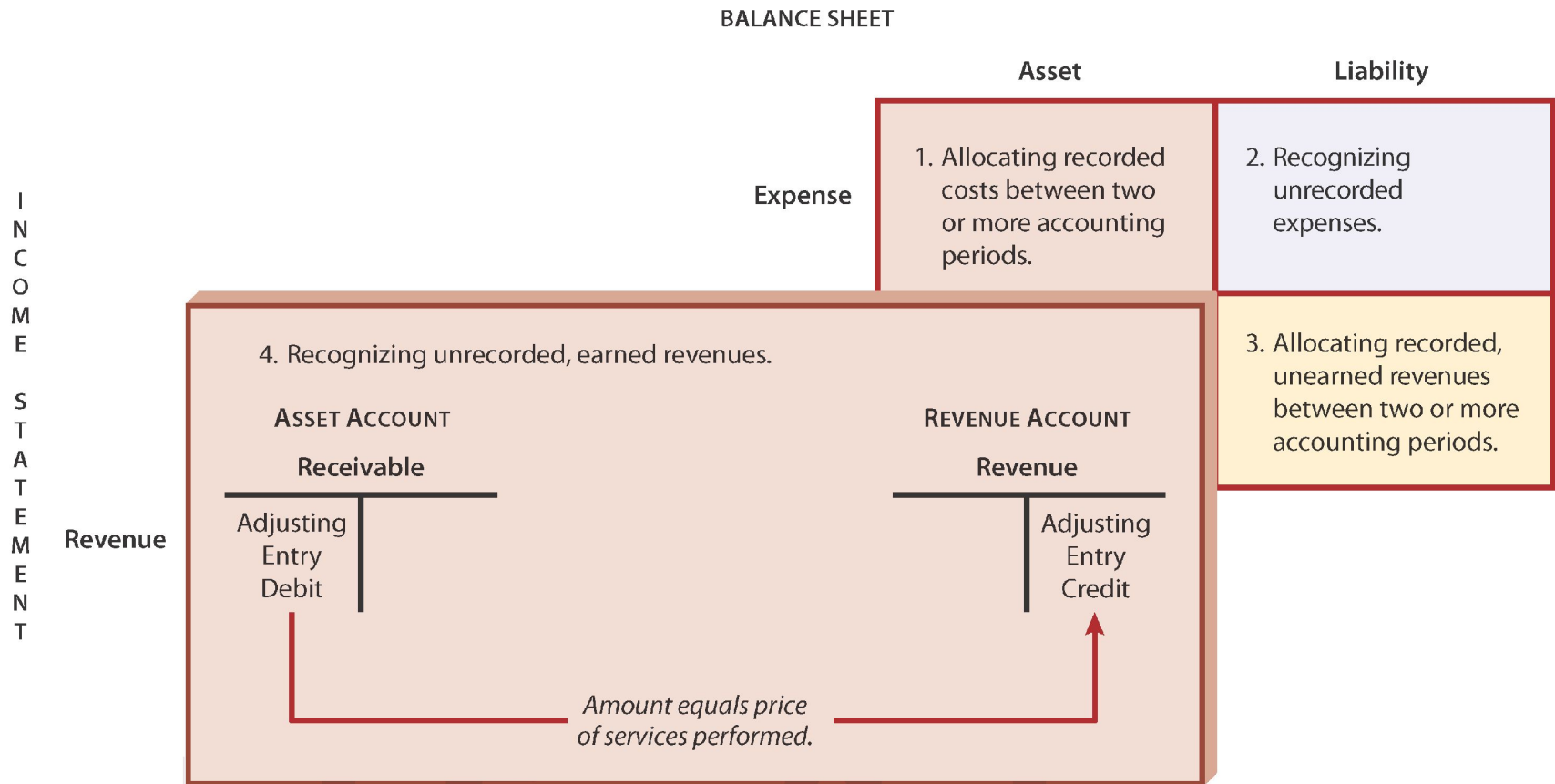
- *Comment:* Unearned Design Revenue now reflects the amount of work still to be performed, \$600.

Type 4 Adjustment:

Recognizing Unrecorded, Earned Revenues

- **Accrued revenues:** Revenues earned but for which no entry has been made in the accounting records and require an adjusting entry.

Adjustment for Unrecorded (Accrued) Revenues



EXAMPLE: Adjustment for Design Revenue (1 of 2)

- **July 31:** Accrual of unrecorded revenue, \$400
- **Analysis:** Accrual of unrecorded revenue
 - ▲ *increases* the stockholders' equity account *Design Revenue* with a credit and
 - ▲ *increases* the asset account *Accounts Receivable* with a debit.

EXAMPLE: Adjustment for Design Revenue (1 of 2)

Application of Double Entry:

Assets				=	Liabilities	+	Stockholders' Equity			
ACCOUNTS RECEIVABLE							DESIGN REVENUE			
July 15	9,600	July 22	5,000					July 10	2,800	
31	400							15	9,600	
Bal.	5,000							31	800	
								31	400	
								Bal.	13,600	

Journal Entry:

		Dr.	Cr.
July 31	Accounts Receivable	400	
	Design Revenue		400

- Comment:** Design Revenue now reflects the total revenue earned during July: \$13,600. On the balance sheet, revenues that have been earned but not recorded are usually combined with accounts receivable. However, some companies prefer to debit an account called Unbilled Accounts Receivable, and others simply flag the transactions in Accounts Receivable as “unbilled.”



Stop & Apply

For each of the items on the left, identify the type of adjusting entry required:

- ___ a. Revenues earned but not yet collected or billed to customers
- ___ b. Interest incurred but not yet recorded
- ___ c. Unused supplies
- ___ d. Costs of plant and equipment accounting periods
- ___ e. Income taxes incurred but not yet recorded

Type 1: Allocating recorded costs between two or more accounting periods

Type 2: Recognizing unrecorded expenses

Type 3: Allocating recorded, unearned revenue between two or more

Type 4: Recognizing unrecorded, earned revenues

SOLUTION

a. Type 4; b. Type 2; c. Type 1; d. Type 1; e. Type 2

Using the Adjusted Trial Balance to Prepare Financial Statements

- An **adjusted trial balance** is prepared by listing all accounts and their balances.
- If the adjusting entries made correctly, the adjusted trial balance will have equal debit and credit totals.

Relationship of the Adjusted Trial Balance to the Income Statement, Balance Sheet, and Statement of Retained Earnings

Creative Designs, Inc. Adjusted Trial Balance July 31, 2011

Cash	\$22,480	
Accounts Receivable	5,000	
Office Supplies	3,660	
Prepaid Rent	1,600	
Office Equipment	16,320	
Accumulated Depreciation— Office Equipment		\$ 300
Accounts Payable		6,280
Unearned Design Revenue		600
Wages Payable		720
Income Taxes Payable		800
Common Stock		40,000
Dividends	2,800	
Design Revenue		13,600
Wages Expense	5,520	
Utilities Expense	680	
Rent Expense	1,600	
Office Supplies Expense	1,540	
Depreciation Expense— Office Equipment		300
Income Taxes Expense	800	
	<u>\$62,300</u>	<u>\$62,300</u>

Creative Designs, Inc. Income Statement For the Month Ended July 31, 2011

Revenues	
Design revenue	13,600
Expenses	
Wages expense	5,520
Rent expense	1,600
Office supplies expense	1,540
Income taxes expense	800
Utilities expense	680
Depreciation expense— office equipment	300
Total expenses	<u>10,440</u>
Net income	<u>\$ 3,160</u>

Creative Designs, Inc. Statement of Retained Earnings For the Month Ended July 31, 2011

Retained earnings, July 1, 2011	—
Net income	3,160
Subtotal	<u>3,160</u>
Less dividends	<u>2,800</u>
Retained earnings, July 31, 2011	<u>\$ 360</u>

Creative Designs, Inc. Balance Sheet July 31, 2011

Assets	
Cash	\$22,480
Accounts receivable	5,000
Office supplies	3,660
Prepaid rent	1,600
Office equipment	16,320
Less accumulated depreciation	<u>300</u>
Total assets	<u>\$48,760</u>
Liabilities	
Accounts payable	\$ 6,280
Unearned design revenue	600
Wages payable	720
Income taxes payable	<u>800</u>
Total liabilities	<u>\$ 8,400</u>
Stockholders' Equity	
Common stock	40,000
Retained earnings	<u>360</u>
Total stockholders' equity	<u>40,360</u>
Total liabilities and stockholders' equity	<u>\$48,760</u>



Stop & Apply

The adjusted trial balance for Carroll Corporation on December 31, 2011, contains the following accounts and balances: Common Stock, \$180; Retained Earnings, \$120; Dividends, \$100; Service Revenue, \$1,100; Rent Expense, \$300; Wages Expense, \$400; Telephone Expense, \$100; and Income Tax Expense, \$50. Compute net income and prepare a statement of retained earnings for the month of December.

SOLUTION

Net income

$$\begin{aligned} &= \$1,100 - \$300 - \$400 \\ &\quad - \$100 - \$50 \\ &= \$1,100 - \$850 \\ &= \$250 \end{aligned}$$

Carroll Corporation

Statement of Retained Earnings

For the Month Ended December 31, 2011

Retained Earnings November 30, 2011	\$120
Net income	<u>250</u>
Subtotal	\$370
Less dividends	<u>100</u>
Retained Earnings, December 31, 2011	\$270

Closing Entries

- **Permanent accounts** (*real accounts*): Carry their end-of-period balances into the next accounting period.
 - Examples: Cash and Accounts Payable
- **Temporary accounts** (*nominal accounts*): Begin each accounting period with a zero balance, accumulate a balance during the period, and cleared by closing entries.
 - Examples: Revenues Earned and Wages Expense
 - **Closing entries**: Journal entries made at the end of an accounting period.

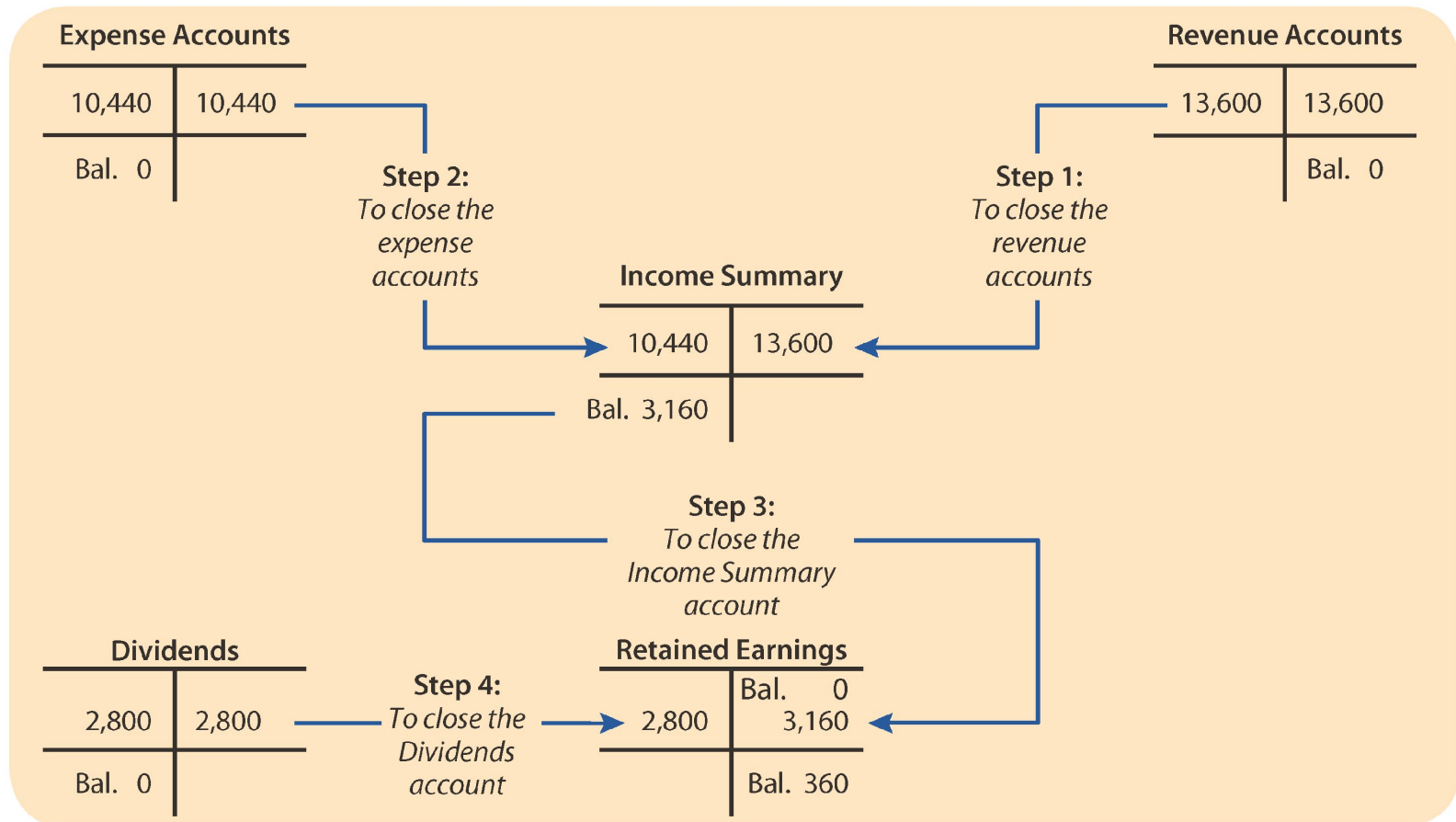
Closing Entries

- Set the stage for the next accounting period.
- Summarize a period's revenues and expenses.
 - **Income Summary account:** Temporary account that summarizes all revenues and expenses for the period.
 - Used only in the closing process—never in the financial statements.

Post-closing trial balance

- Purpose of **post-closing trial balances**
 - determines that all temporary accounts have zero balances
 - double-check that total debits equal total credits
- Contains only balance sheet accounts because the income statement accounts and the Dividends account have all been closed and now have zero balances

Overview of the Closing Process*



*Amounts are for Creative Designs, Inc. See Exhibit S3.1 in the Supplement to Chapter 3.



Stop & Apply

Prepare the necessary closing entries from the following partial adjusted trial balance for MGC Delivery Service, Inc. (except for Retained Earnings, balance sheet accounts have been omitted) and compute the ending balance of Retained Earnings.

**MGC Delivery Service, Inc.
Partial Adjusted Trial Balance
June 30, 2011**

Retained Earnings	\$12,370	
Dividends	\$ 9,000	
Delivery Services Revenue		92,700
Driver Wages Expense	44,450	
Fuel Expense	9,500	
Wages Expense	7,200	
Packing Supplies Expense		3,100
Office Equipment		
Rental Expense	1,500	
Utilities Expense	2,225	
Insurance Expense	2,100	
Interest Expense	2,550	
Depreciation Expense		5,020
Income Taxes Expense		4,500

SOLUTION*Closing entries prepared:*

June 30	Delivery Services Revenue	92,700	
	Income Summary		92,700
	To close the revenue account		
30	Income Summary	82,145	
	Driver Wages Expense		44,450
	Fuel Expense		9,500
	Wages Expense		7,200
	Packing Supplies Expense		3,100
	Office Equipment Rental Expense		1,500
	Utilities Expense		2,225
	Insurance Expense		2,100
	Interest Expense		2,550
	Depreciation Expense		5,020
	Income Taxes Expense		4,500
	To close the expense accounts		
30	Income Summary	10,555	
	Retained Earnings		10,555
	To close the Income Summary account		
	$\$92,700 - \$82,145 = \$10,555$		
30	Retained Earnings	9,000	
	Dividends		9,000
	To close the Dividends account		

Ending balance of Retained Earnings computed:

RETAINED EARNINGS			
June 30	9,000	Beg. Bal.	12,370
		June 30	10,555
		End. Bal.	13,925

Cash Flows from Accrual-Based Information

- Almost every income statement account is related to a balance sheet account.
 - Supplies Expense is related to Supplies.
 - Wages Expense is related to Wages Payable.
 - Design Revenue is related to Unearned Design Revenue.
- Cash inflows and outflows can be analyzed from these relationships.

Determination of Cash Flows from Accrual-Based Information

<i>Type of Account</i>	<i>Potential Payment or Receipt Not Paid or Received</i>				<i>Result</i>
Prepaid Expense	Ending Balance	+	Expense for the Period	– Beginning Balance	= Cash Payments for Expenses
Unearned Revenue	Ending Balance	+	Revenue for the Period	– Beginning Balance	= Cash Receipts from Revenues
Accrued Payable	Beginning Balance	+	Expense for the Period	– Ending Balance	= Cash Payments for Expenses
Accrued Receivable	Beginning Balance	+	Revenue for the Period	– Ending Balance	= Cash Receipts from Revenues



Stop & Apply

Supplies had a balance of \$400 at the end of May and \$360 at the end of June. Supplies Expense was \$550 for the month of June. How much cash was paid for supplies during June? Assume all purchases are for cash.

SOLUTION

Supplies at June 30	\$360
Supplies Expense during June	550
Potential cash payments for supplies	\$910
Less Supplies at May 31	400
Cash payments for supplies during June	\$510